

field of electronics.

"Solar photovoltaic deployment on rooftops, seemingly simple, is potentially revolutionary. It upsets the traditional centralised model of electricity generation and its transmission and distribution. We no longer know what the optimum size, for instance, of the electricity generation solution is. We have assumed that large-scale

coal or nuclear projects, or large dams, are optimal. Are they? Can we have community- or household-level and cost-effective electricity? The innovation potential from solar distributed generation and from the use of micro-grids is enormous. It can fragment and fray the edges of the current industry structure," says professor Mahesh P. Bhavé, visiting professor-Strategy, IIM Kozhikode, setting us thinking on the right lines.

He adds, "Solar is taking off in India, albeit far too slowly. Notice the Railway Minister's budget speech—the railways are rightly emphasising solar deployment. They can hasten the process."

Signifying the rising demand for solar energy, we now see semiconductor companies across the world taking a huge interest in improving photovoltaic (PV) technology and offering better solutions for the solar energy sector. "Today, to cater to the needs of the solar energy market, there are a range of new semiconductor products and solutions, with high performance and high efficiency, dedicated specifically to photovoltaic applications, for the major grid-tied architectures and for solar battery applications," says Mongia.

Consider, for example, STMicroelectronics' portfolio of photovoltaic ICs, which includes cool bypass switches designed to improve the reliability of panel electronics, and DC-DC converters with built-in maximum power point tracking (MPPT), which maximise power conversion of solar panels independent of temperature



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and the amount of solar irradiation. The new 'cool bypass' devices minimise internal energy losses when routing high-efficiency solar power, by combining efficient power switching and intelligent control in a single chip to compensate for the variable effects of hotspots and shadows on the solar panel surface. These devices can recover and save up to 1 per cent of the energy produced and normally lost through conventional bypass diodes.

"And, we are not standing still. Even our cool bypass devices have now moved onto the next generation of 'iceberg' devices. ST has introduced several innovative technologies that improve efficiency in solar-power converters, including our proprietary MDmesh and STripFETM VII DeepGATETM power MOSFETs that ensure ultra-low-loss performance, trenchgate field-stop IGBTs, and the second generation of silicon-carbide (SiC) Schottky diodes for minimising switching losses and improving thermal performance in solar-power systems, where capturing every fractional improvement in energy efficiency is valuable. ST's trials, using the company's latest 1200V silicon carbide diodes, have delivered a 2 per cent increase in overall inverter yield, even when operating at a high load and high frequency," says Mongia.

Micro-inverter-based solar grids

Current-generation solar devices implement the MPPT algorithm to

maximise the energy produced from each solar panel at any panel temperature and radiation condition. This maximises the overall efficiency of the photovoltaic system. This may be implemented traditionally or using a distributed approach.

"The traditional grid-tier architecture of photovoltaic sys-

tems concentrates all the electronics in the central inverter. This is the centralised approach. The trend today is to move towards the distributed approach where the electronics is partially or fully distributed close to each panel. This is beneficial in terms of global system energy production, reliability, communications and monitoring," says Mongia.

Mongia explains the two most common options:

String and central-inverter-based architecture. This represents the most common power conversion system for grid-connected applications, and generally comprises three main functional blocks: The junction box in each solar panel provides the key bypass functionality. The string combiner box provides the protection and monitors the solar panel strings. The inverter provides the MPPT to adapt the impedance that the panel views at its output to obtain maximum power transfer, the DC-DC power conversion stage to adapt voltage levels, and the DC-AC power conversion stage to correctly shape the current and voltage waveforms, and to connect to the AC grid.

Micro-inverter-based architecture. This represents a fully distributed photovoltaic grid-connected system in which all the electronics is moved close to each panel. This has two main functional blocks: the micro-inverter provides the MPPT, the complete power conversion, the connectivity and AC grid connection, while the data